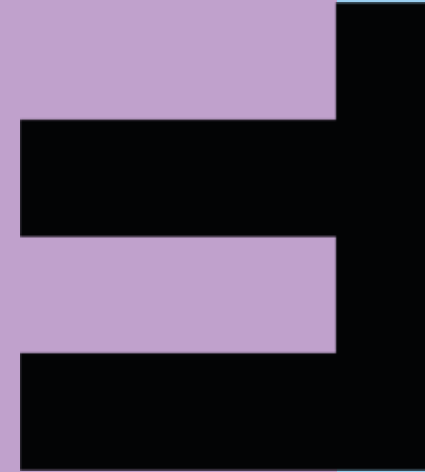


FHV

Vorarlberg University
of Applied Sciences



Technical Information Systems

7. SQL – Exercises with the AdventureWorks Databases

Exercises – Multiple tables

1. List all students with the lecturers of their classes

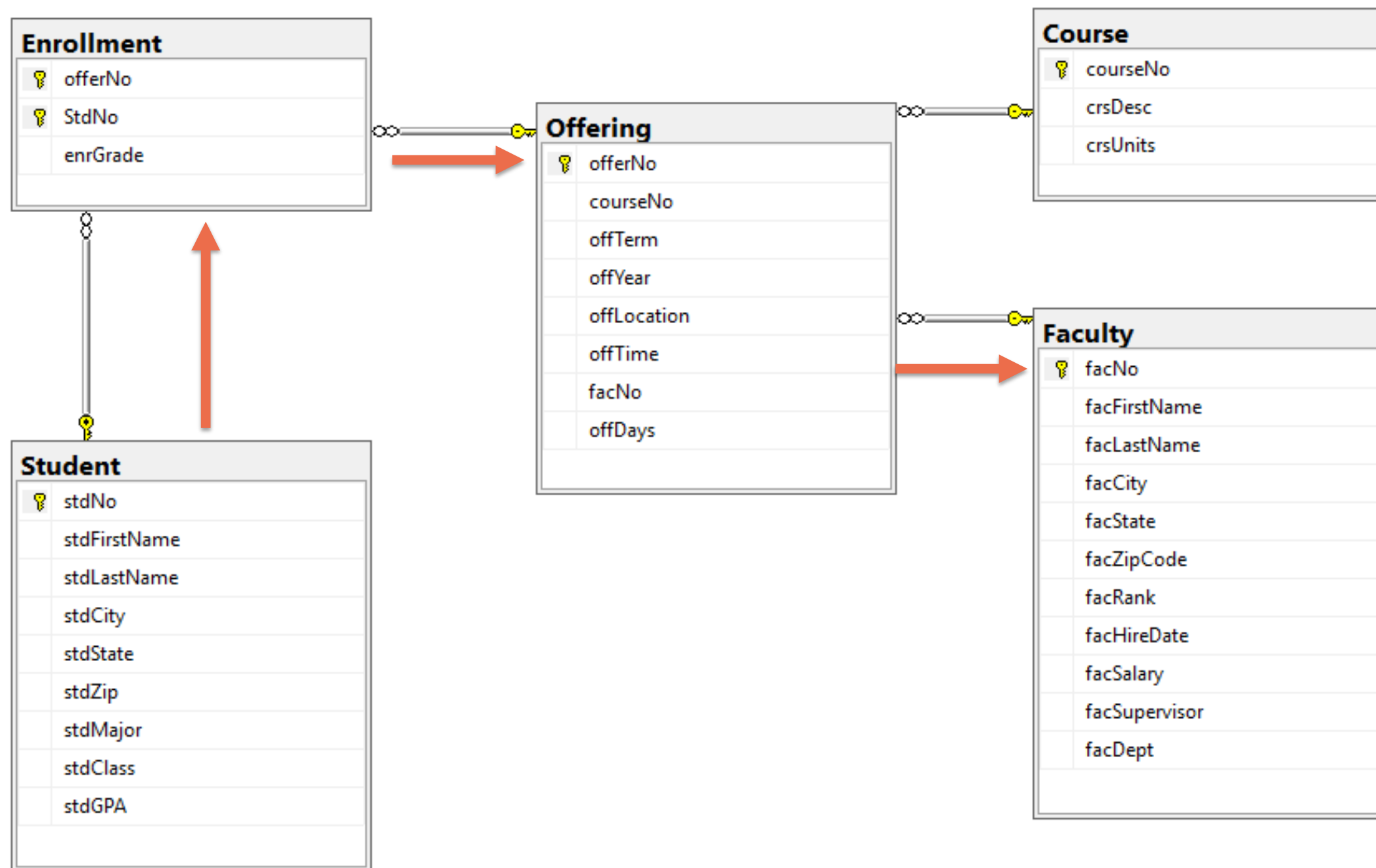
Extend the code as follows (one after the other)

2. Don't allow for repeated rows
3. Sort by student name and lecturer name
4. Count the classes a student has with a lecturer
5. Sort by number of classes held together starting with the highest number

	stdFirstName	stdLastName	facFirstName	facLastName	(No column name)	
1	HOMER	WELLS	LEONARD	VINCE	3	
2	LUKE	BRAZZI	LEONARD	VINCE	2	
3	MARIAH	DODGE	JULIA	MILLS	2	
4	TESS	DODGE	LEONARD	VINCE	2	
5	CANDY	KENDALL	JULIA	MILLS	2	
6	ROBERTO	MORALES	LEONARD	VINCE	2	

Exercises – Multiple tables

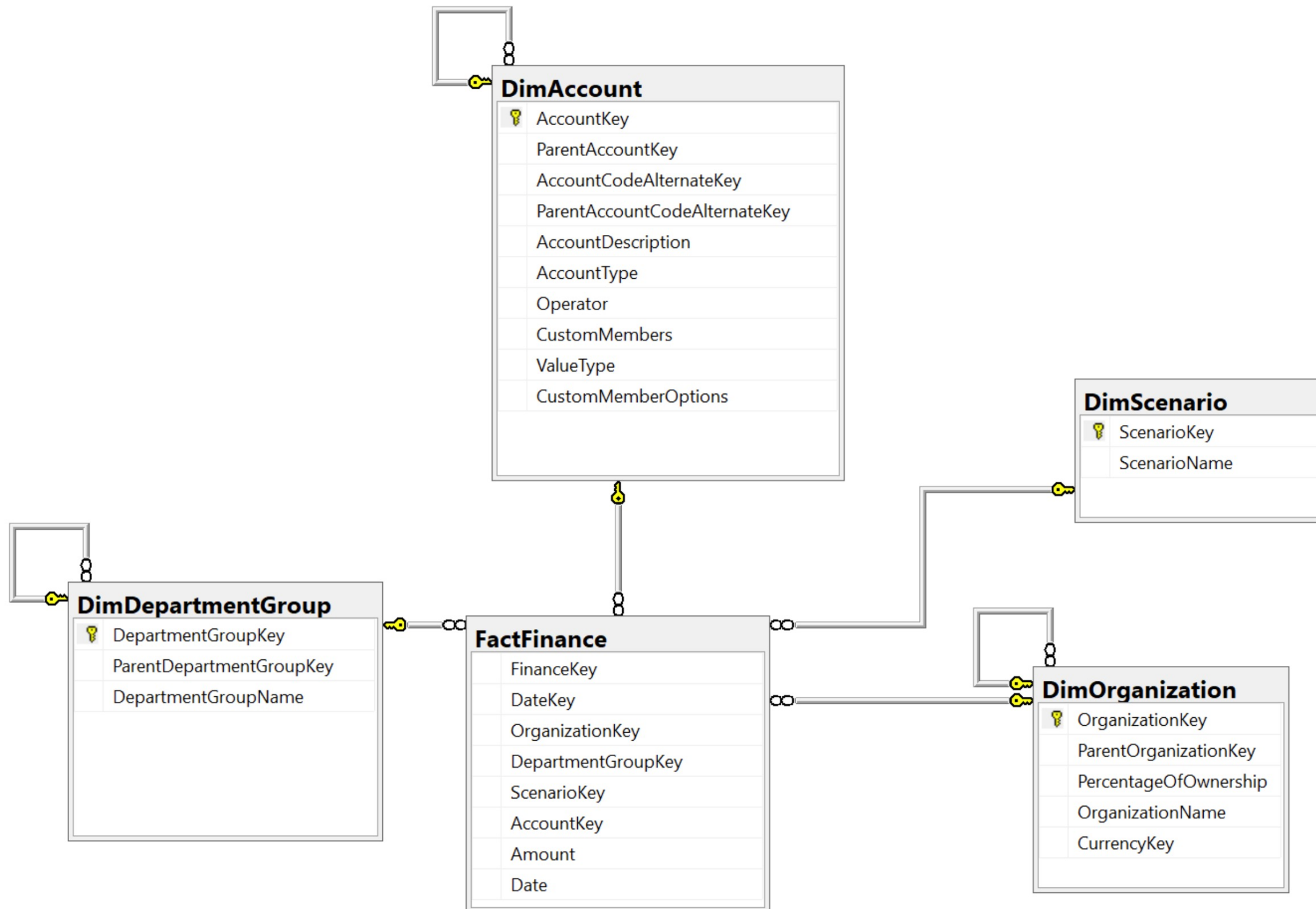
- List all students with the lecturers of their classes



Exercises – Multiple tables



AdventureWorksDW2012



AdventureWorksDW2012

- List all amounts in FactFinance with their account description
- Format the amount column as US-Currency

	amount	AccountDescription
1	\$22,080.00	Salaries
2	\$20,200.00	Salaries
3	\$2,000.00	Payroll Taxes
4	\$2,208.00	Payroll Taxes
5	\$1,546.00	Employee Benefits



- List all Departments with the sum of their amounts

	DepartmentGroupName	(No column name)
1	Corporate	\$21,053,924.61
2	Executive General and Administration	\$38,237,453.45
3	Inventory Management	\$84,542,533.92
4	Manufacturing	\$1,627,440.25
5	Quality Assurance	\$870,742.17
6	Research and Development	\$221,314,887.40
7	Sales and Marketing	\$990,993,430.90



1

1. How does data become information and information become knowledge?
2. What distinguishes Object data from Meta data?
3. What happens, if one omits one attribute of a candidate key?
Why?

2

- 4. Is there always a key candidate in a record? Why?
- 5. Why is it not advisable to index all attributes of a table?

What is the advantage of indexing some?

- 6. If a foreign key is defined unique – what kind of relationship (x:y) does it belong to?
- 7. What is the difference between a Key and an Index?

3

7. Find trivial superkey, superkey and key candidates in the following table (assuming that the given data reflects all possible variations):

Artist	Title	Published
The X	Trouble	2012
The X	Greatest Hits	2013
MoneyMaker	Greatest Hits	2011
Tralala	Sing along	2011

4a

Regarding the table below

8. How many records are in the table? How many attributes?

9. Which redundancies are in the table?

Which consequences follow from these redundancies?

Client	Client_Phone	Client_Adress	Zip	Consultant	Contact-date	Duration	Hourly_Rate	Consultant_Knowhow
Emil Schmidt	0719-5661 2876	Appleroad 23, Suntown	9394	Helena Maier	21.3.2016	1 hour	€50,00	Finance, Accounting
Hans Müller	0092-6002 1404	Bananaplace 33, Moontown	6238	Ingo Fox	30.4.2016	4 hours	€45,00	Marketing, Strategy
Johanna Schulte	0572-6413 9798	Pearstreet 543, Cloudvalley	4181	Helena Maier	1.5.2016	3 hours	€50,00	Finance, Accounting
Markus Schulz	0510-3319 2045	Lemonplace 1, Startown	4122	Ingo Fox	3.5.2016	1 hour	€45,00	Marketing, Strategy
Hans Müller	0092-6002 1404	Bananaplace 33, Moontown	6238	Ingo Fox	12.5.2016	1 hour	€45,00	Marketing, Strategy
Johanna Schulte	0572-6413 9798	Pearstreet 543, Cloudvalley	4181	Helena Maier	15.5.2016	1 hour	€50,00	Finance, Accounting
Hans Müller	0092-6002 1404	Bananaplace 33, Moontown	6238	Ingo Fox	30.5.2016	2 hours	€45,00	Marketing, Strategy

4b

10. Which entities can you identify in the table?

Which attributes belong to which entity?

Client	Client_Phone	Client_Adress	Zip	Consultant	Contact-date	Duration	Hourly_Rate	Consultant_Knowhow
Emil Schmidt	0719-5661 2876	Appleroad 23, Suntown	9394	Helena Maier	21.3.2016	1 hour	€50,00	Finance, Accounting
Hans Müller	0092-6002 1404	Bananaplace 33, Moontown	6238	Ingo Fox	30.4.2016	4 hours	€45,00	Marketing, Strategy
Johanna Schulte	0572-6413 9798	Pearstreet 543, Cloudvalley	4181	Helena Maier	1.5.2016	3 hours	€50,00	Finance, Accounting
Markus Schulz	0510-3319 2045	Lemonplace 1, Startown	4122	Ingo Fox	3.5.2016	1 hour	€45,00	Marketing, Strategy
Hans Müller	0092-6002 1404	Bananaplace 33, Moontown	6238	Ingo Fox	12.5.2016	1 hour	€45,00	Marketing, Strategy
Johanna Schulte	0572-6413 9798	Pearstreet 543, Cloudvalley	4181	Helena Maier	15.5.2016	1 hour	€50,00	Finance, Accounting
Hans Müller	0092-6002 1404	Bananaplace 33, Moontown	6238	Ingo Fox	30.5.2016	2 hours	€45,00	Marketing, Strategy

4c

11. How are the entities related to each other?

Which entity is superordinate to the others?

12. What happens if you delete record 2, 5 and 7?

	Client	Client_Phone	Client_Adress	Zip	Consultant	Contact-date	Duration	Hourly_Rate	Consultant_Knowhow
1	Emil Schmidt	0719-5661 2876	Appleroad 23, Suntown	9394	Helena Maier	21.3.2016	1 hour	€50,00	Finance, Accounting
2	Hans Müller	0092-6002 1404	Bananaplace 33, Moontown	6238	Ingo Fox	30.4.2016	4 hours	€45,00	Marketing, Strategy
3	Johanna Schulte	0572-6413 9798	Pearstreet 543, Cloudvalley	4181	Helena Maier	1.5.2016	3 hours	€50,00	Finance, Accounting
4	Markus Schulz	0510-3319 2045	Lemonplace 1, Startown	4122	Ingo Fox	3.5.2016	1 hour	€45,00	Marketing, Strategy
5	Hans Müller	0092-6002 1404	Bananaplace 33, Moontown	6238	Ingo Fox	12.5.2016	1 hour	€45,00	Marketing, Strategy
6	Johanna Schulte	0572-6413 9798	Pearstreet 543, Cloudvalley	4181	Helena Maier	15.5.2016	1 hour	€50,00	Finance, Accounting
7	Hans Müller	0092-6002 1404	Bananaplace 33, Moontown	6238	Ingo Fox	30.5.2016	2 hours	€45,00	Marketing, Strategy

5

- 13. What is a n:m relation? How do you build it in a database?
- 14. Why are there different data models to describe a planned (or existing) data base?
- 15. What is an entity?
- 16. Explain simple, complex, stored, derived and multi-valued attributes.
- 17. The database should be able to store different credit card numbers (max 3) for one person. Name variants to model this.
- 17.b What would change if there were people with joint accounts?

6

- 18. Does normalization eliminate all data redundancies? Why?
- 19. In the formal relational model, a relation is defined as a set of tuples. Explain what this means. What consequences does this have, e.g. for the definition of keys?

7 Database Modelling

- Rules for Database Modelling:
 - To exercise, create a conceptual schema using Chen Diagram (first) and an implementation schema with Martin Notation with all needed attributes (that the narrative states) and keys.
 - The narrative is complete. Do not interpret any extra meanings into the story. If you discover ambiguities in the narrative, make up reasonable assumptions to complete the story and state the assumptions made.
 - Use natural keys if feasible.
 - If not specially stated, you do not have to name data formats.
 - The solution must follow rules for 3rd normal form

7.1

1. A database to keep track of business trips of employees

- Each employee has an employee-ID, a name (first, last) and belongs to a department.
- For each trip information about start and end town and start- and end-date is needed and the employee(s) taking part in that trip
- The travel expenses of a trip are related to an employee and a department that is charged for it.
- All expenses (e.g. flight, taxi, restaurant) must to be listed separately

7.2

2. Library

- A library has books and borrowers
- Each book comes from a specific publisher and is written by one or more authors. It is available in the library in one or more copies.
- Each borrower can borrow at maximum 20 books.
- Each book can be borrowed for a specific time. This time can be extended by 4 weeks twice (if there is no reservation).

Extension

- If all copies of a book are checked out, borrowers can request a reservation.

7.2 – Solution

Remarks:



7.3

3. A bigger sport happening

- The happening consists of multiple events (each event taking place at only one complex).
- There are a number of sports complexes where the events can happen (even at the same time).
- Every complex has multiple areas.
- During an event, there can be multiple competitions (maximal one at each areas).
- Each sport can take place in only one areas.
- Each sport needs several equipment pieces.
- The equipment can be used by several sports.
- Each piece of equipment is located at one complex and can't be moved to other complexes.

7.4

4. AIRLINE relational database – Create the Chen diagram only:

- Each flight is identified by a flight number, and consists of one or more flight legs with leg numbers 1, 2, 3, and so on.
- Each flight leg has scheduled arrival and departure times, airports, and one or more leg instances – one for each date on which the flight travels.
- For each flight leg instance, number of seats and seat reservations are kept, as is the airplane used on the leg and the actual arrival and departure times and airports.
- An airplane has an airplane id and is of a particular airplane type.
- Reservations are made for a specific customer and for single legs.
- The different airplane types can only land on specific airports. These need to be listed in the database.
- An airport is identified by an airport code.

c.f. (Elmasri; Navathe 2010, P.80)

8. Normalisierung – Eine Bücherliste

Tabelle *books*

<u>bookNo</u>	title	edition	authors	publisher	publCity	year
1	An Introduction to Database ...	3	Date	Addison-Wesley	Reading	1981
2	The Relational Model ...	1	Codd	Addison-Wesley	Reading	1990
3	Time and Relational Theory ...	2	Date, Darwen, Lorentzos	Morgan Kaufmann	Burlington	2014
4	Database Design ...	2	Date	Apress	New York	2019
...						

Abbildung 4.42 Ausgangstabelle einer Bücherliste

Beispiel aus: [Kofl22:191-193]

8. Normalisierung – Eine Bücherliste – 1. NF

Tabelle <i>books</i>							
<u>bookNo</u>	title	edition	author	<u>authorRank</u>	publisher	publCity	year
1	An Introduction to Database ...	3	Date	1	Addison-Wesley	Reading	1981
2	The Relational Model ...	1	Codd	1	Addison-Wesley	Reading	1990
3	Time and Relational Theory ...	2	Date	1	Morgan Kaufmann	Burlington	2014
3	Time and Relational Theory ...	2	Darwen	2	Morgan Kaufmann	Burlington	2014
3	Time and Relational Theory ...	2	Lorentzos	3	Morgan Kaufmann	Burlington	2014
4	Database Design ...	2	Date	1	Apress	New York	2019
...							

Abbildung 4.43 Die Bücher in der ersten Normalform

8. Normalisierung – Eine Bücherliste – 2. NF?

Tabelle *books*

<u>bookNo</u>	title	edition	publisher	publCity	year
1	An Introduction to Database ...	3	Addison-Wesley	Reading	1981
2	The Relational Model ...	1	Addison-Wesley	Reading	1990
3	Time and Relational Theory ...	2	Morgan Kaufmann	Burlington	2014
4	Database Design ...	2	Apress	New York	2019
...					

Tabelle *authors*

<u>authNo</u>	<u>bookNo</u>	authorRank	author
1	1	1	Date
2	2	1	Codd
1	3	1	Date
3	3	2	Darwen
4	3	3	Lorentzos
1	4	1	Date
...	...		

8. Normalisierung – Eine Bücherliste – 2. NF

Tabelle *books*

<u>bookNo</u>	title	edition	publisher	publCity	year
1	An Introduction to Database ...	3	Addison-Wesley	Reading	1981
2	The Relational Model ...	1	Addison-Wesley	Reading	1990
3	Time and Relational Theory ...	2	Morgan Kaufmann	Burlington	2014
4	Database Design ...	2	Apress	New York	2019
...					

Tabelle *authors*

<u>authNo</u>	author
1	Date
2	Codd
3	Darwen
4	Lorentzos
...	

Tabelle *authorRank*

<u>authNo</u>	<u>bookNo</u>	authorRank
1	1	1
2	2	1
1	3	1
3	3	2
4	3	3
1	4	1

aber noch Probleme mit der dritten Normalform.

8. Normalisierung – Eine Bücherliste – 3. NF

Tabelle <i>books</i>				
<u>bookNo</u>	title	edition	publNo	year
1	An Introduction to Database ...	3	1	1981
2	The Relational Model ...	1	1	1990
3	Time and Relational Theory ...	2	2	2014
4	Database Design ...	2	3	2019
...				

Tabelle <i>authors</i>	
<u>authNo</u>	author
1	Date
2	Codd
3	Darwen
4	Lorentzos
...	

Tabelle <i>authorRank</i>		
<u>authNo</u>	<u>bookNo</u>	authorRank
1	1	1
2	2	1
1	3	1
3	3	2
4	3	3
1	4	1
...	...	

Tabelle <i>publishers</i>		
<u>publNo</u>	publisher	publCity
1	Addison-Wesley	Reading
2	Morgan Kaufmann	Burlington
3	Apress	New York
...		

Abbildung 4.46 Nun entsprechen alle Tabellen auch der dritten Normalform.

9. Normalisierung - Autoverkauf

Kunden-name	Adresse	Auto-marke	Typ	Serien-nummer	Verkäufer	Datum
Meier	Planeten-weg 7	VW	Golf	123456	Schmid	23.4.08
		Opel	Astra	345678	Plüss	7.8.08
Müller	Altstadt 12	VW	Golf	388721	Frey	17.6.08
Steffen	Gartenstr. 7	VW	Bora	222245	Schmid	15.7.08
Steffen	Augasse 12	Audi	A6	122154	Frey	13.11.08
		Opel	Antara	445321		
					Schenk	

9. Normalisierung – Autoverkauf 1. NF

Geschäftsdaten

KNr	Kun- den- name	Adresse	ANr	Auto- Marke	Typ	Serien- num- mer	VNr	Ver- käufer	Datum
1	Meier	Planetenweg 7	1	VW	Golf	123456	1	Schmid	23.4.08
1	Meier	Planetenweg 7	2	Opel	Astra	345678	2	Plüss	7.8.08
2	Müller	Altstadt 12	3	VW	Golf	388721	3	Frey	17.6.08
3	Steffen	Gartenstr. 7	4	VW	Bora	222245	1	Schmid	15.7.08
4	Steffen	Augasse 12	5	Audi	A6	122154	3	Frey	13.11.08
			6	Opel	Antara	445321			
							4	Schenk	

9. Normalisierung – Autoverkauf 2. NF

Kunden

KNr	Kundenname	Adresse
1	Meier	Planetenweg 7
2	Müller	Altstadt 12
3	Steffen	Gartenstr. 7
4	Steffen	Augasse 12

Verkäufe

KNr	ANr	Datum	VNr	Verkäufer
1	1	23.4.08	1	Schmid
1	2	7.8.08	2	Plüss
2	3	17.6.08	3	Frey
3	4	15.7.08	1	Schmid
4	5	13.11.08	3	Frey

Verkäufer
Schenk?

Autos

ANr	Automarke	Typ	Seriennummer
1	VW	Golf	123456
2	Opel	Astra	345678
3	VW	Golf	388721
4	VW	Bora	222245
5	Audi	A6	122154
6	Opel	Antara	445321

9. Normalisierung – Autoverkauf 3. NF

Kunden

KNr	Kundenname	Adresse
1	Meier	Planetenweg 7
2	Müller	Altstadt 12
3	Steffen	Gartenstr. 7
4	Steffen	Augasse 12

Autos

ANr	Automarke	Typ	Seriennummer
1	VW	Golf	123456
2	Opel	Astra	345678
3	VW	Golf	388721
4	VW	Bora	222245
5	Audi	A6	122154
6	Opel	Antara	445321

Verkäufe

KNr	ANr	Datum	VNr
1	1	23.4.08	1
1	2	7.8.08	2
2	3	17.6.08	3
3	4	15.7.08	1
4	5	13.11.08	3

Verkäufer

VNr	Verkäufer
1	Schmid
2	Plüss
3	Frey
4	Schenk